C1
Data Representation

Outline

1. Bits and Bytes
2. Representation of Character Sets
   a. ASCII
   b. Unicode
3. Representation of Data Types
   - Numeric, Boolean, Character, String
4. Representation of Numbers
5. Representation of Images, Sound and Video
   (not included in Syllabus)

1. Bits and Bytes

- Values are stored inside a computer as a series of 0s and 1s



Binary digit or bit

1. Bits and Bytes

- How big is a Byte?
- Depend on the computer (hardware) architecture
  - Early computers by CDC Corp : 6-bit byte
  - Early computers by BB&N : 10-bit byte
  - Today's computers : 8-bit byte



1. Bits and Bytes

- Byte Size determines the maximum number of possible values that can be stored.
  - 1 bit : 0 or 1 → 2 = 2^1 values
  - 2 bits : 00, 01, 10 or 11 → 4 = 2^2 values
  - 3 bits : 000, 001, 010, 011, 100, 101, 110 or 111 → 8 = 2^3 values
  - 4 bits : … → 2^4 = 16 values
  - …
  - 8 bits : … → 2^8 = 256 values
  - 16 bits : … → 2^{16} = 65,536 values
  - 32 bits : … → 2^{32} = 4,294,967,296 values

1. Bits and Bytes

- The bits themselves have no intrinsic meaning, the interpretation of the values is determined by the way the hardware and software use the bits

2. Representation of Character Sets

- Using a byte for a character, we have 256 codes, enough to represent each of the characters on a standard keyboard

01000001

2. Representation of Character Sets

Computer 1	Computer 2
01000001→ ‘A’	11001001→ ‘A’

- These two computers cannot communicate with each other because they uses different Character Representation Sets.

2. Representation of Character Sets a. ASCII

- 1960s – agreed a standard set of codes
- American Standard Code for Information Interchange (ASCII)
- "Ask-key"

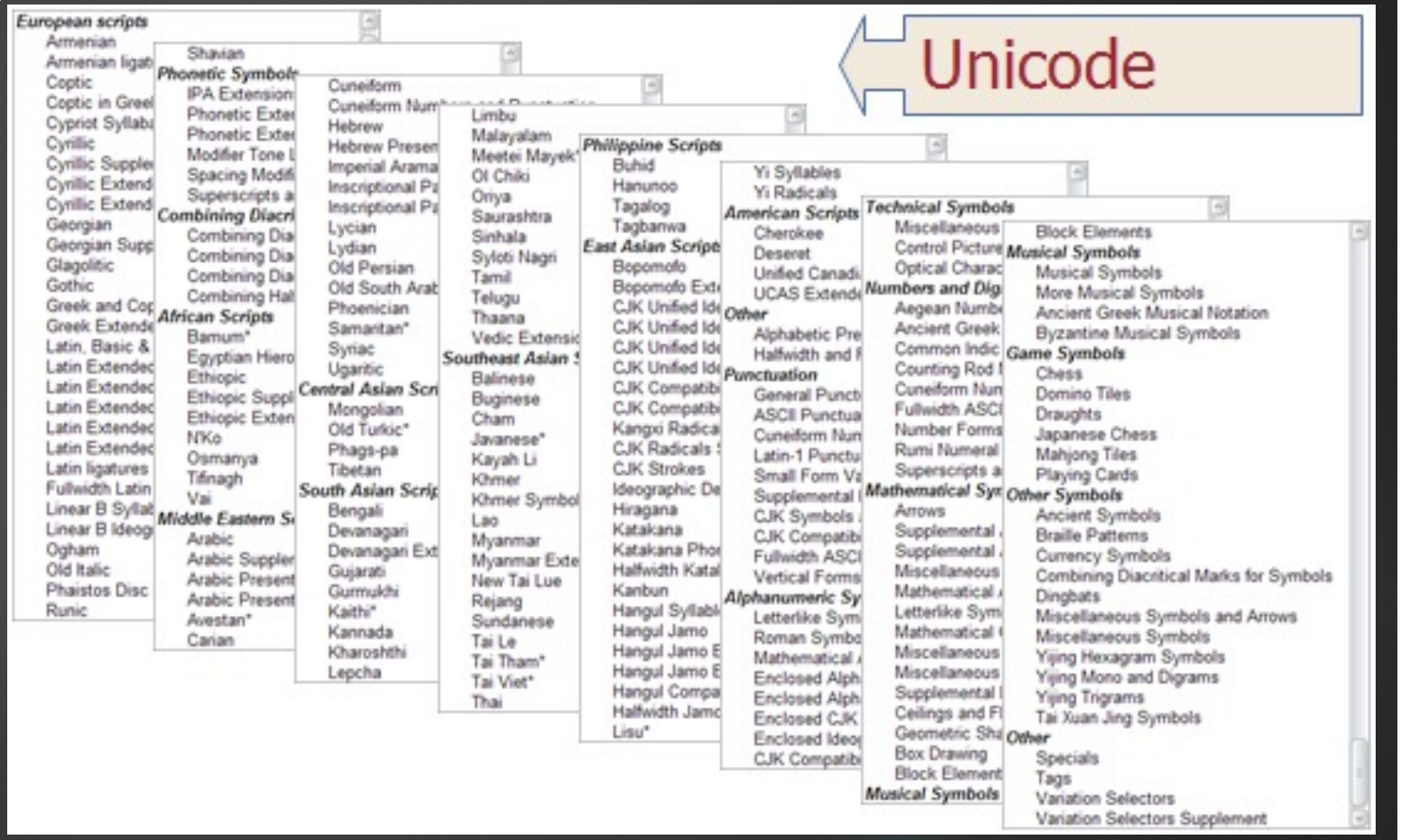
2. Representation of Character Sets
a. ASCII

- Uses 7 bits to represent each character and the 8th bit as a means of checking the rest of the 7 bits.

  (we will learn this under the topic '*Parity Check*')

- 128 characters can be represented in the standard ASCII character set.

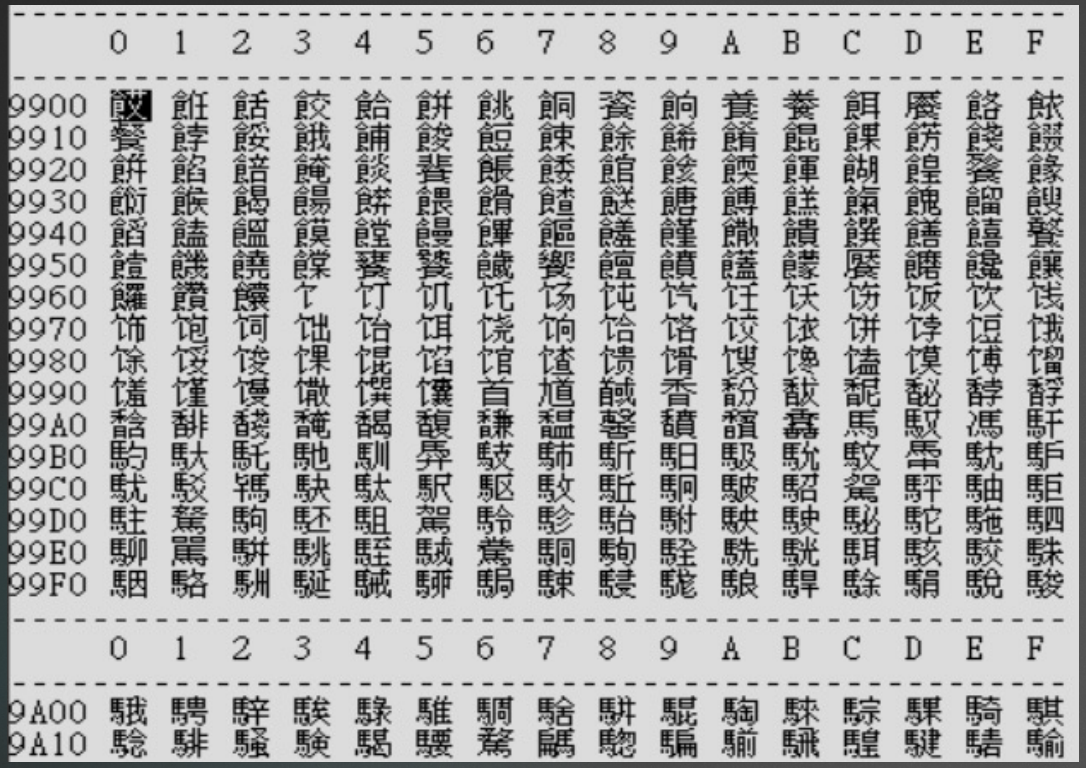
2. Representation of Character Sets
b. Unicode



b. Unicode



- More recent
- 16-bit code (2 bytes)
- Can represent over 65,000 characters
  - All the characters used by the world's languages
- Widely used to handle documents that is required to be written in different languages
- Allows localisation of software – standard software can be adapted for use in different cultures by modifying the layout and features



	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
9900	餘	餕	餑	餃	餉	餅	餚	餖	餐	餉	養	養	餌	饜	餉	餓
9910	餐	餅	餚	餵	餼	餾	餽	餾	餘	餘	餾	餛	餕	餚	餞	餞
9920	餅	餚	餡	餺	餽	餻	餵	餼	館	餘	餾	餜	餑	餉	餾	餾
9930	餚	餵	餛	餾	饒	餻	餺	餽	餽	餾	餾	餵	饌	餅	餾	餅
9940	餾	餼	餞	餾	餼	饕	餾	餾	餾	餾	餾	餼	饜	餚	餾	餾
9950	餾	餽	餾	餼	饌	餻	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
9960	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
9970	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
9980	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
9990	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
99A0	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
99B0	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
99C0	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
99D0	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
99E0	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾
99F0	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾	餾

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
9A00	騫	騬	騪	騷	騸	騮	騶	騷	騹	騽	騾	驀	驂	驄	驏	驑
9A10	驗	驞	驕	驗	驘	驤	驚	驪	驁	驅	驒	驟	驍	驤	驥	驗

- Chinese Character Set – thousands of different characters

0	■	32		64	@	96	`	128	■	160		192	À	224	à
1	■	33	!	65	A	97	a	129	■	161	¡	193	Á	225	á
2	■	34	"	66	B	98	b	130	‚	162	¢	194	Â	226	â
3	■	35	#	67	C	99	c	131	ƒ	163	£	195	Ã	227	ã
4	■	36	$	68	D	100	d	132	„	164	¤	196	Ä	228	ä
5	■	37	%	69	E	101	e	133	…	165	¥	197	Å	229	å
6	■	38	&	70	F	102	f	134	†	166	¦	198	Æ	230	æ
7	■	39	'	71	G	103	g	135	‡	167	§	199	Ç	231	ç
8	■	40	(	72	H	104	h	136	ˆ	168	¨	200	È	232	è
9	■	41	)	73	I	105	i	137	‰	169	©	201	É	233	é
10	■	42	*	74	J	106	j	138	Š	170	ª	202	Ê	234	ê
11	■	43	+	75	K	107	k	139	‹	171	«	203	Ë	235	ë
12	■	44	,	76	L	108	l	140	Œ	172	¬	204	Ì	236	ì
13	■	45	-	77	M	109	m	141	■	173	-	205	Í	237	í
14	■	46	.	78	N	110	n	142	■	174	®	206	Î	238	î
15	■	47	/	79	O	111	o	143	■	175	¯	207	Ï	239	ï
16	■	48	0	80	P	112	p	144	■	176	°	208	Ð	240	ð
17	■	49	1	81	Q	113	q	145	‘	177	±	209	Ñ	241	ñ
18	■	50	2	82	R	114	r	146	’	178	²	210	Ò	242	ò
19	■	51	3	83	S	115	s	147	“	179	³	211	Ó	243	ó
20	■	52	4	84	T	116	t	148	”	180	´	212	Ô	244	ô
21	■	53	5	85	U	117	u	149	•	181	µ	213	Õ	245	õ
22	■	54	6	86	V	118	v	150	–	182	¶	214	Ö	246	ö
23	■	55	7	87	W	119	w	151	—	183	·	215	×	247	÷
24	■	56	8	88	X	120	x	152	˜	184	¸	216	Ø	248	ø
25	■	57	9	89	Y	121	y	153	™	185	¹	217	Ù	249	ù
26	■	58	:	90	Z	122	z	154	š	186	º	218	Ú	250	ú
27	■	59	;	91	[	123	{	155	›	187	»	219	Û	251	û
28	■	60	<	92	\	124	|	156	œ	188	¼	220	Ü	252	ü
29	■	61	=	93	]	125	}	157	■	189	½	221	Ý	253	ý
30	■	62	>	94	^	126	~	158	■	190	¾	222	Þ	254	þ
31	■	63	?	95	_	127	■	159	Ÿ	191	¿	223	ß	255	ÿ

■ Indicates that this character isn't supported by Windows.
TT Indicates that this character is available only in TrueType fonts.

- ANSI Set – Graphical symbols, lines and shapes

3. Representing Data Types – Numerical Data

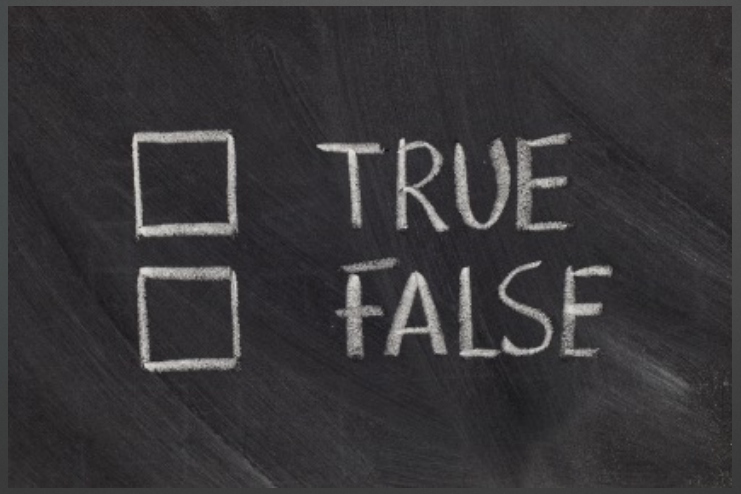
- Integers
- Negative numbers
- Real numbers

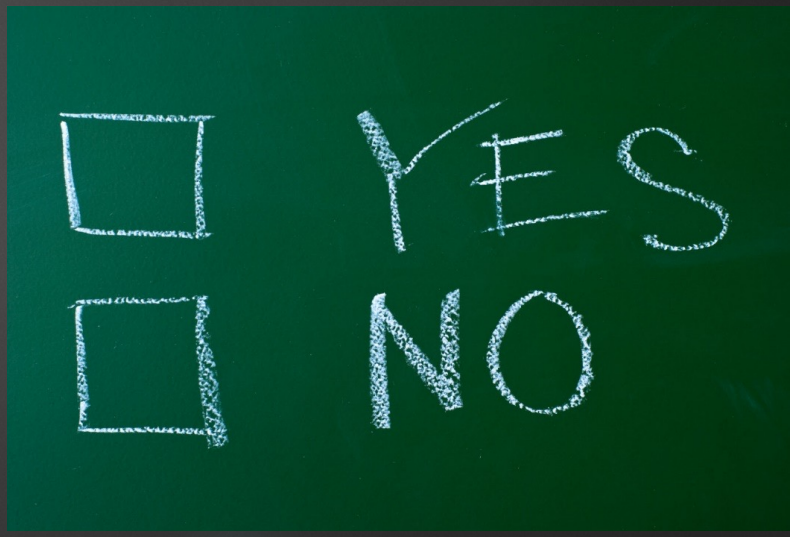
- Size: either 1, 2 or 4 bytes
  - Larger number of bytes can store larger numbers
    - 1 byte → 0 to 255
    - 2 bytes → 0 to 65,535

3. Representing Data Types – Boolean Data

- Data with two states
- 0 or 1
- Size: 1 bit







3. Representing Data Types – Character & String Data

- Character → single letter, digit or symbol
- Size: usually 1 byte

- String → sequence of characters stored together
- Size: depends on the number of characters

4. Representation of Numbers

1. Unsigned (Positive) Integers
2. Signed Integers (not in Syllabus)
    1. Sign-Magnitude
    2. One's Complement
    3. Two's Complement
3. Floating Point Numbers (not in Syllabus)

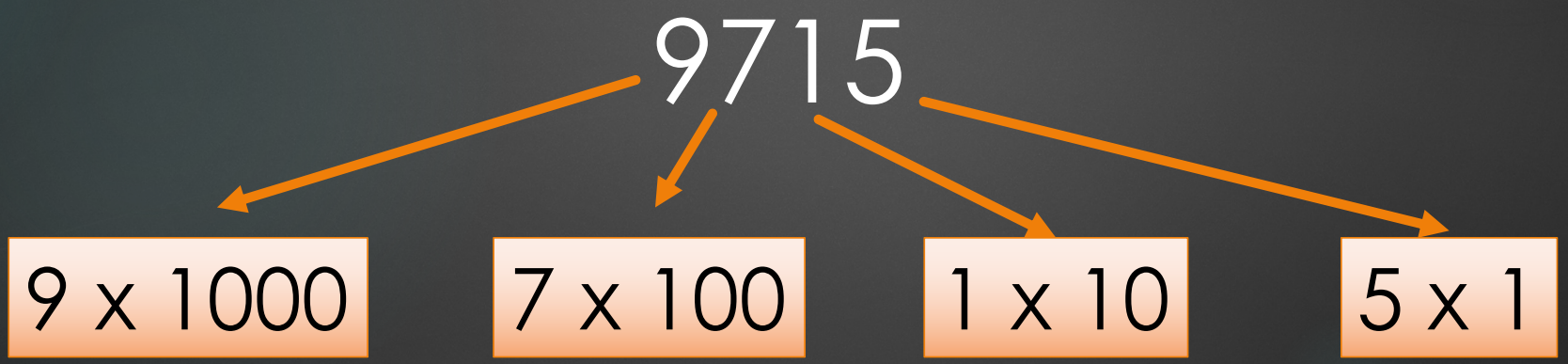
3. Representation of Positive Integers

3. Representation of Positive Integers

Hindu-Arabic	Roman
1	I
5	V
10	X
50	L
100	C
500	D
1 000	M

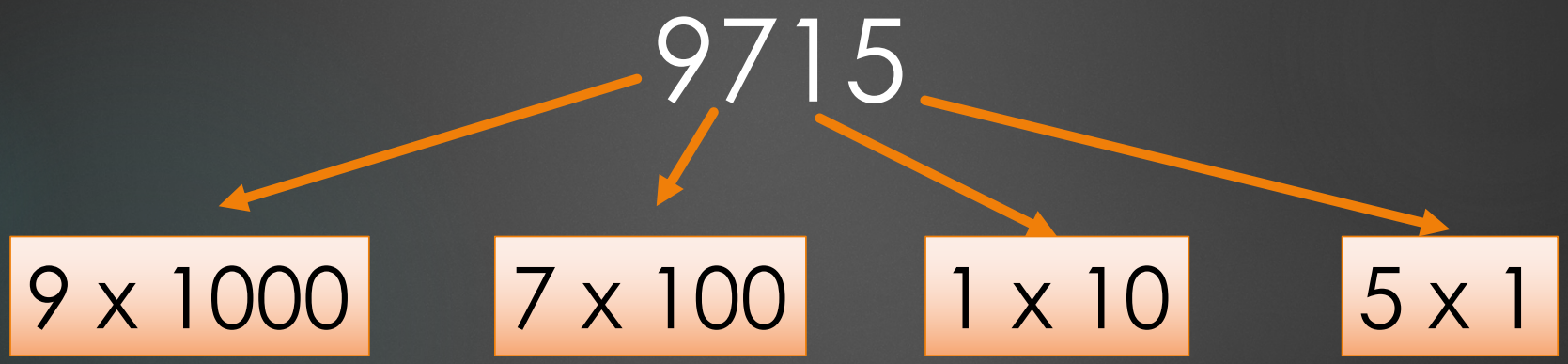
Positional Weighted System

- The position of a digit in a number carries different weight
- Example (Base 10):
  - Digits in base 10 = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}



Formal definition {base(radix) **10**}

$$\left(a_n a_{n-1} \ldots a_0\right)_{10} = a_n \times 10^n + a_{n-1} \times 10^{n-1} + \ldots + a_0 \times 10^0$$



$$9715_{10} = 9 \times 10^3 + 7 \times 10^2 + 1 \times 10^1 + 5 \times 10^0$$

How about other bases?

Base	"Digits" in the system	Example number	Value in base 10
10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9	1011_{10}	$1 \times 10^3 + 1 \times 10^1 + 1 \times 10^0 = 1011_{10}$
2	0, 1	1011_2	$1 \times 2^3 + 1 \times 2^1 + 1 \times 2^0 = 11_{10}$
3	0, 1, 2	1011_3	$1 \times 3^3 + 1 \times 3^1 + 1 \times 3^0 = 31_{10}$
8	0, 1, 2, 3, 4, 5, 6, 7	1011_8	$1 \times 8^3 + 1 \times 8^1 + 1 \times 8^0 = 521_{10}$
16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F	1011_{16}	$1 \times 16^3 + 1 \times 16^1 + 1 \times 16^0 = 4113_{10}$

Common Bases

- Base 10 = Denary
  - Decimal number formed by digits (0, 1, …, 9)
- Base 2 = Binary
  - Binary number formed by bits (0, 1)
- Base 8 = Octal
  - Octal number formed by octals (0, 1, …, 7)
- Base 16 = Hexadecimal
  - Hexadecimal number formed by hexadigits (0, 1, …, 9, A, B, … , F)

Exercise:
Convert Base-*R* to Decimal

- 1101_2
- 572_8
- $2A_{16}$
- 341_5

Decimal to Binary Conversion

- Method 1
  - Repeated Division by 2
- Method 2
  - Sum of Weight

Method 1: Repeated Division by 2

- Use successive division by 2 until the quotient is 0:
- The remainders form the answer:
  - The first remainder is the least significant bit (LSB)
  - The last remainder is the most significant bit (MSB)



Exercise: Convert Decimal to Binary (Method 1)

89	
24	
100	
73	
127	

Method 2: Sum of Weights

- Determine the set of binary weights whose sum is equal to the decimal number

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

- $9_{10} = 8 + 1 = 2^3 + 2^0 = 1001_2$
- $18_{10} = 16 + 2 = 2^4 + 2^1 = 10010_2$
- $58_{10} = 32 + 16 + 8 + 2 = 2^5 + 2^4 + 2^3 + 2^1 = 111010_2$

Exercise: Convert Decimal to Binary (Method 2)

89		128	64	32	16	8	4	2	1
24		128	64	32	16	8	4	2	1
100		128	64	32	16	8	4	2	1
73		128	64	32	16	8	4	2	1
127		128	64	32	16	8	4	2	1

Decimal to Base-R Conversion

- Method 1 is a "sure-fire" method
  - Just repeatedly divide by R

- Method 2 still works, BUT…
  - It is usually harder as you now need to test for multiples of a weight

	3^2	3^1	3^0
	9	3	1
19	0, 1, 2	0, 1, 2	0, 1, 2

Exercise: Convert the Decimal 89 (89_{10}) to Base- R

Base	Answer
5	
8	
16	
10	89

Base-K to Base-J Conversion

- In general,
  - Use the decimal system as the bridge
- Base-K → Base-10 → Base-J

Exercise: Convert Binary to Octal

Binary	Octal
1010111010	
1111101100	
1011000001	
1000111110	
1001000101	
1101101010	

Exercise: Convert Binary to Hexadecimal

Binary	Hexadecimal
1010111010	
1111101100	
1011000001	
1000111110	
1001000101	
1101101010	

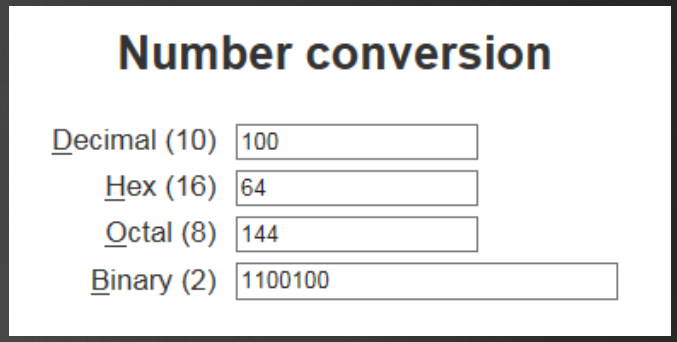
Exercise: Octal ←→ Hexadecimal

Octal	Hexadecimal
537	
123	
65	
	AAA
	BOB
	FACE

Online Tools :

- Number Conversion

  https://coderstoolbox.net/number/



- IEEE-754 Floating Point Converter

  https://www.h-schmidt.net/FloatConverter/IEEE754.html